

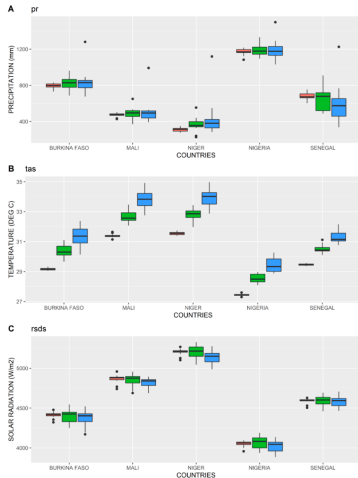
Lessons from the field - Applications of Genomic Offset to African and Mediterranean crops

Philippe Cubry - philippe.cubry@ird.fr

SSMPG2025, Aussois, 11 sept 2025

UMR DIADE-IRD

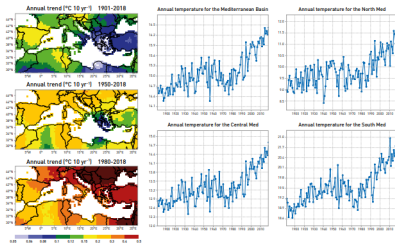
WEST AFRICA AND MEDITERRANEAN REGIONS FACING CLIMATE CHANGE



Climate change predictions in the Sahelian region

- First variable : Monsoon onset predicted over West-African countries
- 156 derived variables in three classes (Temperatures, Precipitations, Solar radiations)
- Several climatic models and greenhouse gaz emission scenarii

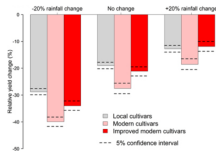
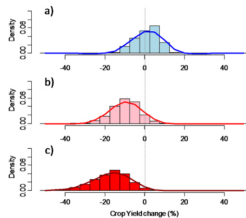
WEST AFRICA AND MEDITERRANEAN REGIONS FACING CLIMATE CHANGE



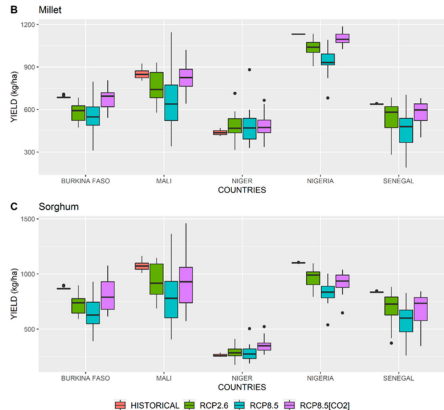
MedECC report, 2024

Climate change at higher rate in the Mediterranean - *"annual mean temperatures are now 1.54°C above the 1860-1890 level for land and sea areas, i.e. 0.4°C more than the global average change (high confidence)." and this change will be more pronounced than in other regions - "Future regional average warming will exceed the global mean value by 20% on an annual basis and 50% in summer (high confidence)."*

PEARL MILLET - A STAPLE CROP AT RISK



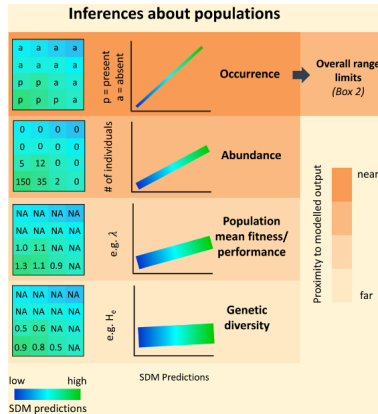
Sultan et al, 2013



Defrance et al, 2020

From crop-growth models calibrated with a restricted number of varieties -> need from a framework to assess the risk for traditional landraces (mal)adaptation

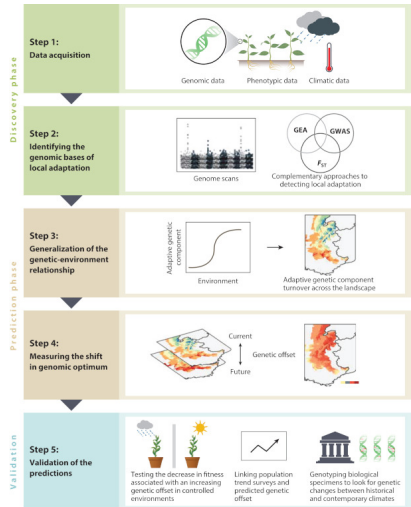
WHY GENOMIC OFFSET ?



Lee-Yaw et al, 2021

-> we need to consider genomic diversity in the models

CAN GENOMIC OFFSET BE LINKED TO YIELD PREDICTION ?



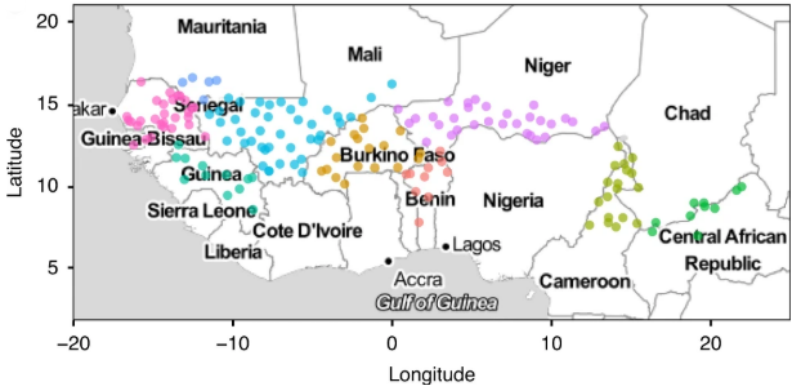
Capblancq T, et al. 2020.
Annu. Rev. Ecol. Syst. 51:345-69



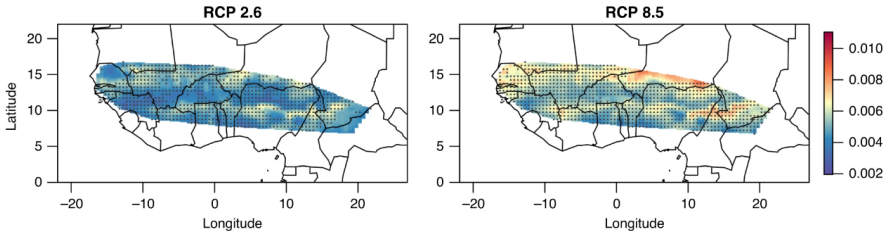
Rhoné et al, 2020, using Gradient Forest on
pearl millet (*Pennisetum glaucum*)



A FIRST STUDY



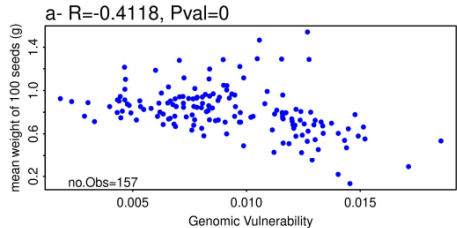
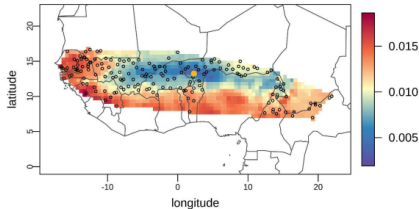
- 173 local populations, 100 accessions/pop
- pool-seq capture experiment (targeting exome, c.a. 140k SNPs)
- 156 climatic variables computed relatively to monsoon onset locally estimated (collab. with agro-climatologists)



Rhoné et al, 2020

- Mean GO value from 17 climate models
- Points indicate GO values highly consistent ($CV < 50\%$) between the different models

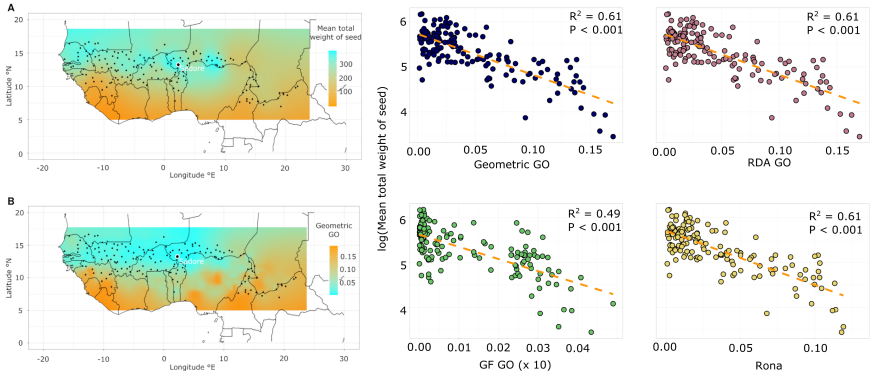
- Common garden experiment to link offset estimate with loss in yield - pretty good and highly significant correlation



Rhoné et al, 2020

Note : initial GF implementation didn't took into account genetic structure (trials using Latent Factors as constant variables in the model improved the correlations)

- Clément Gain and Olivier reanalysis of the data with LEA v3

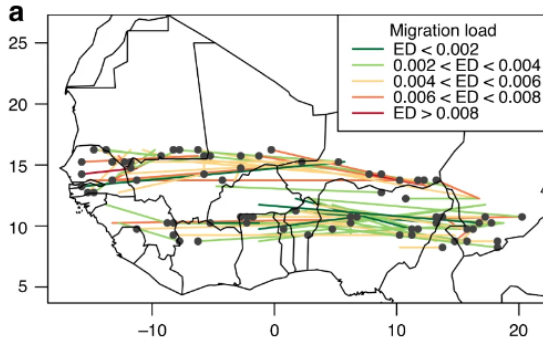


Gain et al, 2023

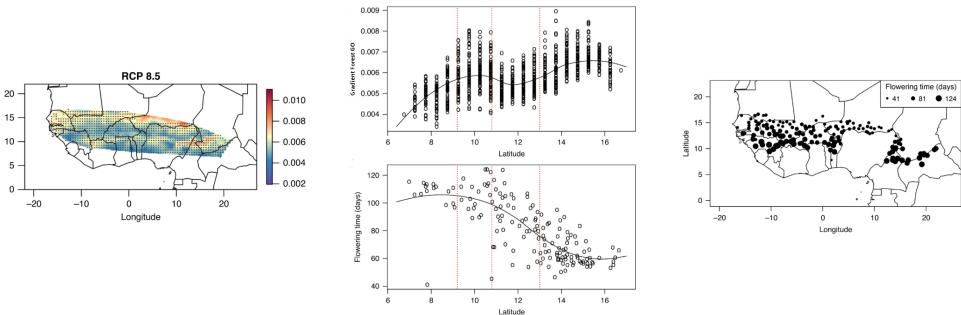
Greatly improve the correlations between GO and loss in yield in the common garden

DOING MORE WITH THE OFFSETS

Using the GO framework to identify putative migrations for highly (mal)-adpated landraces - Identifying origins that could minimize the offset for these highly threatened points



DOING MORE WITH THE OFFSETS



Rhoné et al, 2020

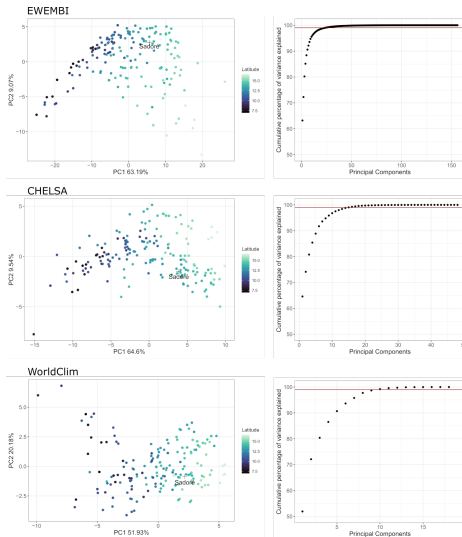
- Assessing biological significance of the offset - link with key agronomic traits
- SNP identified by GWAS on FT were better predicted by the climate predictors identified in the GF model than the genome-wide average



Mind your climate data - Post-doc T. Kaczmarek

- Effect of different climate dataset on the GO predictions/loss in yield on pearl millet
- Commonly used databases WorldClim/CHELSA which provide 19 "bio-climatic" variables (+ extended set in Chelsa)
- Comparison with our climate dataset (156 variables computed relatively to a locally predicted monsoon onset - derived from collaboration with agro-climatologists)

BENCHMARKING ENVIRONMENTAL VARIABLES

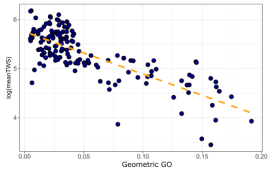


- PCA to identify PCs that explained up to 99% of variance explained to be included in the model
- Identifying the environmental location of the common garden

BENCHMARKING ENVIRONMENTAL VARIABLES

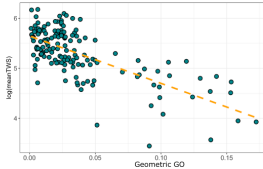
EWEMBI dataset

Adjusted $R^2 = 0.603$; $P < 2.2 \cdot 10^{-16}$



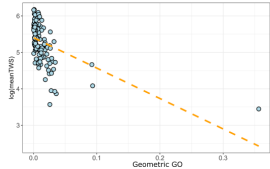
CHELSEA dataset

Adjusted $R^2 = 0.480$; $P < 2.2 \cdot 10^{-16}$



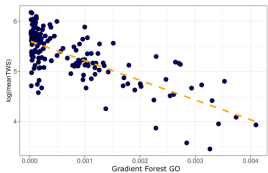
WorldClim dataset

Adjusted $R^2 = 0.249$; $P = 2.6 \cdot 10^{-11}$



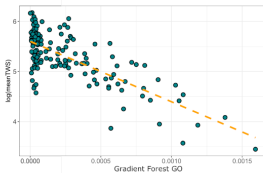
EWEMBI dataset

Adjusted $R^2 = 0.536$; $P < 2.2 \cdot 10^{-16}$



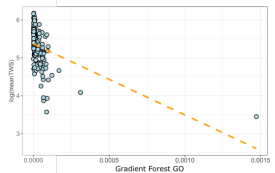
CHELSEA dataset

Adjusted $R^2 = 0.548$; $P < 2.2 \cdot 10^{-16}$



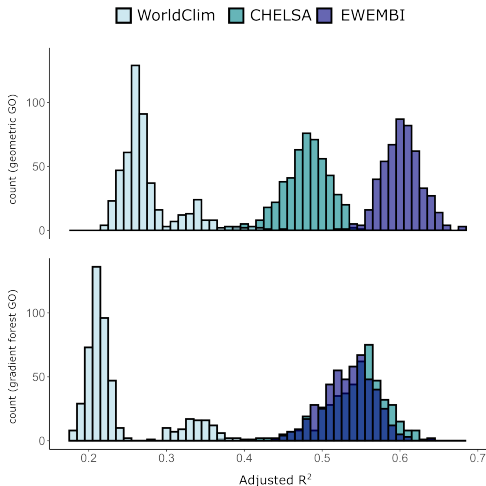
WorldClim dataset

Adjusted $R^2 = 0.199$; $P = 4.2 \cdot 10^{-09}$



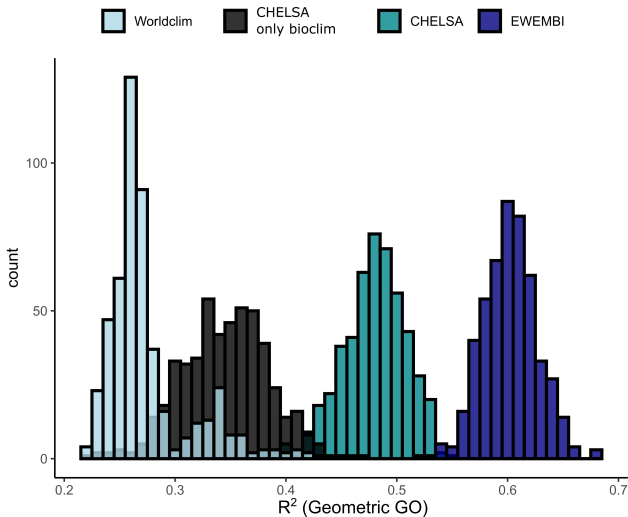
- Both geometric GO and GF assessed
- WordClim consistently performed poorly

BENCHMARKING ENVIRONMENTAL VARIABLES



- Resampling 90% of accessions - 500 reps
- Best combination = geometric GO + EWEMBI

BENCHMARKING ENVIRONMENTAL VARIABLES



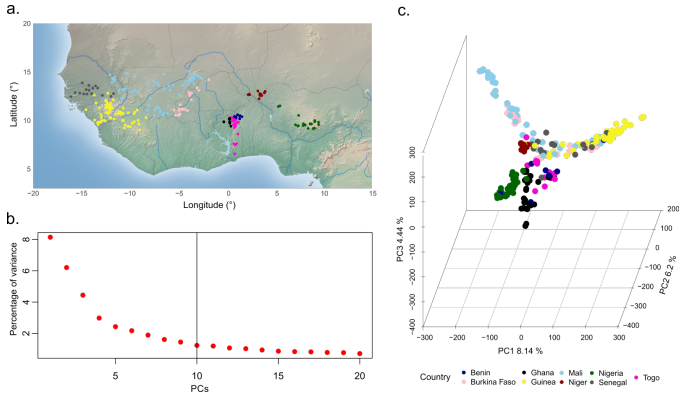
FONIO - THE "HUNGRY RICE" - GENOMIC OFFSET



PhD T. Kaczmarek, Kaczmarek et al, *in prep*



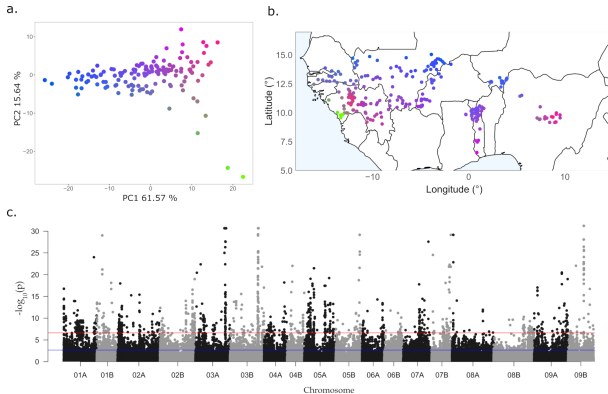
FONIO - THE "HUNGRY RICE" - GENOMIC OFFSET



Kaczmarek et al, in prep

- More than 250 landraces from west Africa
- Complete genome re-sequencing data - c.a. 200k SNP
- 156 climate variables

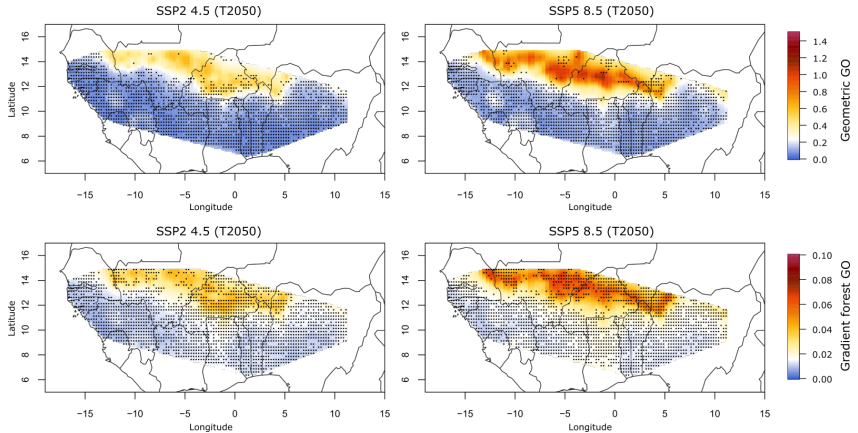
FONIO - THE "HUNGRY RICE" - GENOMIC OFFSET



Kaczmarek et al, in prep

- PCA on climate variables
- Retaining PCs up to 99% of variance explained
- GEA conducted using LFMM

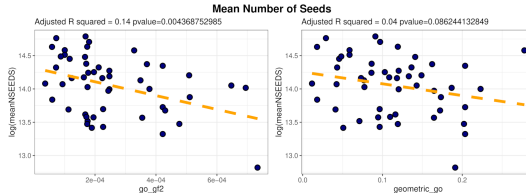
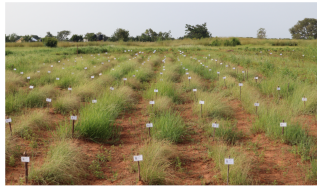
FONIO - THE "HUNGRY RICE" - GENOMIC OFFSET



Kaczmarek et al, in prep

- Different models generally agree but are more inconsistent at high GO

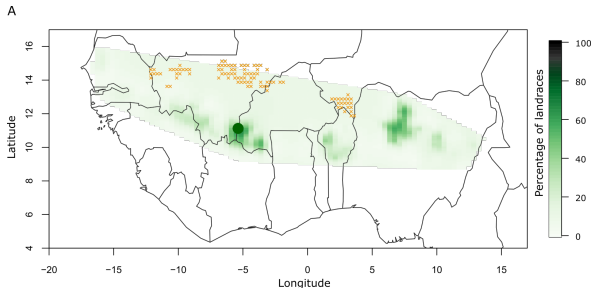
FONIO - THE "HUNGRY RICE" - GENOMIC OFFSET



Kaczmarek et al, in prep

- Common garden trials
- Lower but significant correlation between GO estimate and relative loss of yield (with GF) - depends on the trait surveyed

Doing more with the Offsets - Identifying conservation places



Kaczmarek et al, in prep

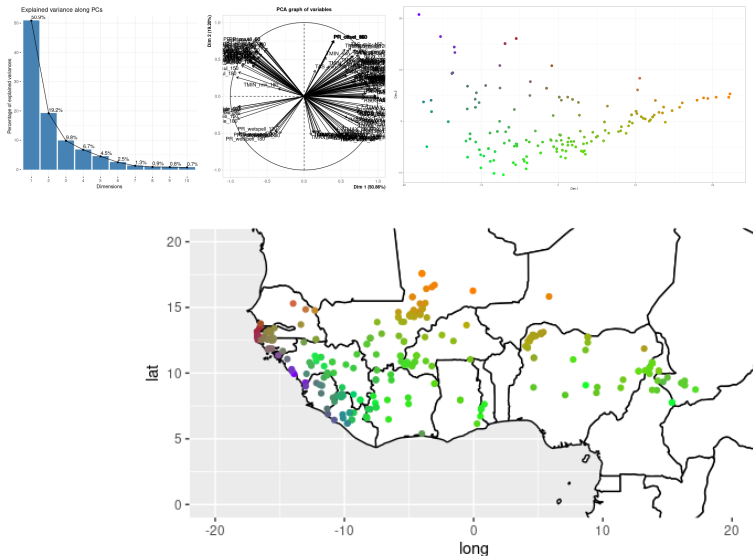
- Identify pixels with 25% highest GO values
- Test their forward GO values and identify the pixels with lowest values
- Retain pixels that maximised the number of landraces



M2 J. Lokossou

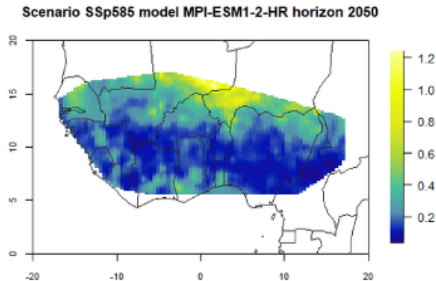
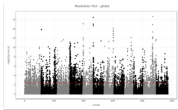


THE AFRICAN RICE - GENOMIC OFFSET



264 genotypes, c.a. 100K SNP, 156 climate variables

THE AFRICAN RICE - GENOMIC OFFSET



- GEA on PCs explaining up to 99% of the environmental variance
- Geometric GO
- High vulnerabilities found at the northern and southern edges of the range

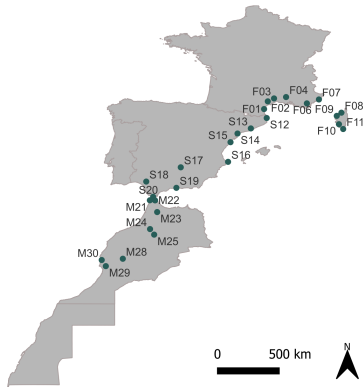


PhD L. Zunino, Local adaptation and GEA on oleasters

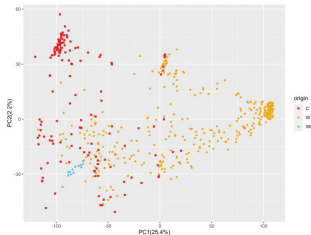


ANOTHER FIELD, THE OLIVE TREE

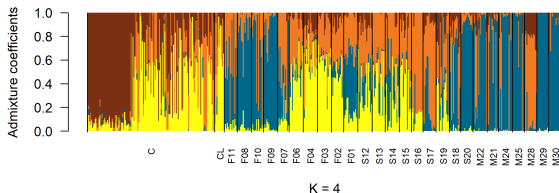
- A complex domestication history that might have blurred local adaptation -> First step : identification of "genuine" wild trees from the western Mediterranean basin
- A 13° latitudinal gradient
- 26 sampled pops
- 359 spontaneous trees
- One wild eastern population
- Cultivated varieties
- Target sequencing experiment (all predicted genes in a ref genome)
- c.a. 300k SNP



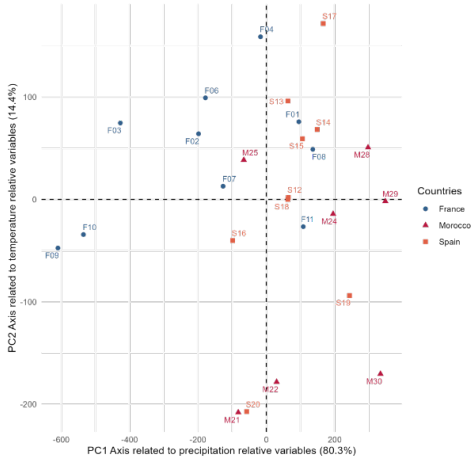
ANOTHER FIELD, THE OLIVE TREE



- Extensive gene flow
- Still signal for a western genepool (esp in Morocco and Corsica)

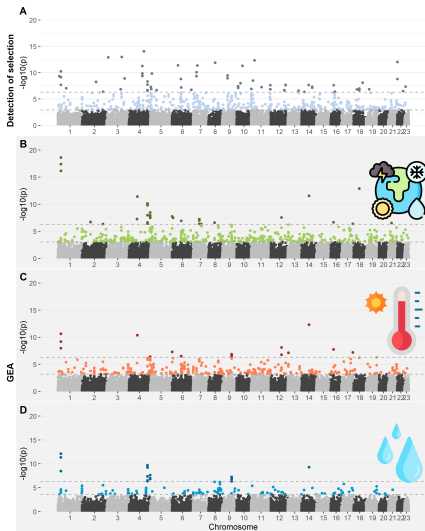


ANOTHER FIELD, THE OLIVE TREE



- PCA on CHELSA bioclim variables for sampled populations
- Contrasting environment along the latitudinal gradient
- PC1 mainly represent precipitation-related variables, PC2 temperature-related ones

ANOTHER FIELD, THE OLIVE TREE



- Signal for local adaptation on wild trees
- GEA made on different aspects of the climate
- Some polymorphisms are shared, others specific to one climate aspect

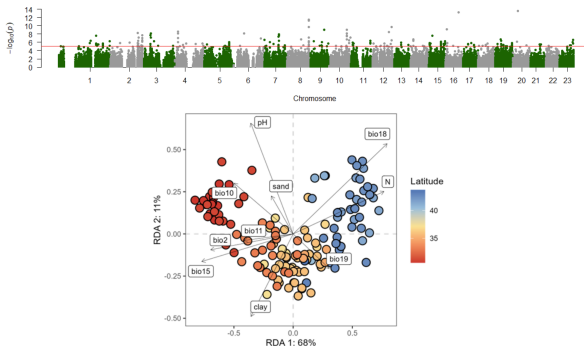
ANOTHER FIELD, THE OLIVE TREE



Post-doc L. Rocchetti, using GO to assess where olive cultivars should thrive



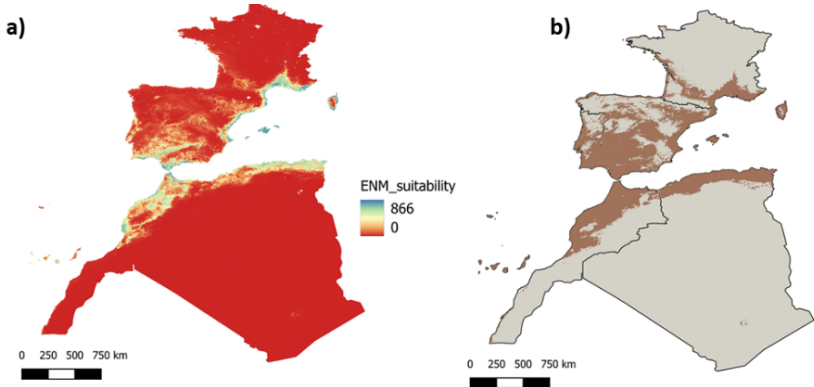
ANOTHER FIELD, THE OLIVE TREE



Rocchetti et al, in prep

- GEA (LFMM) on 10 chosen climate (CHELSA) and soil (SoilGrids) variables only based on wild genotypes
- RDA to model the adaptive landscape based on adaptive variants shared with cultivated

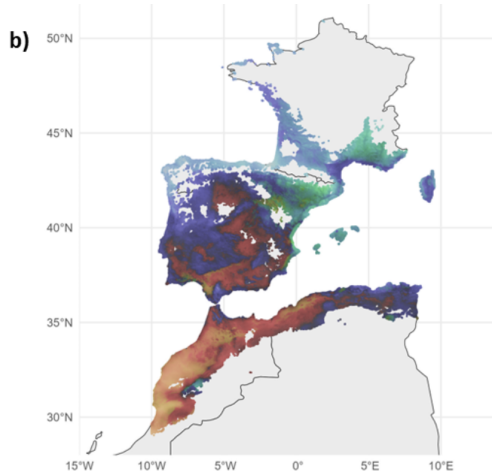
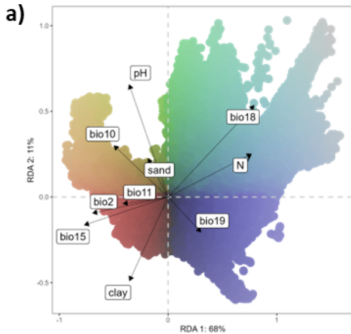
ANOTHER FIELD, THE OLIVE TREE



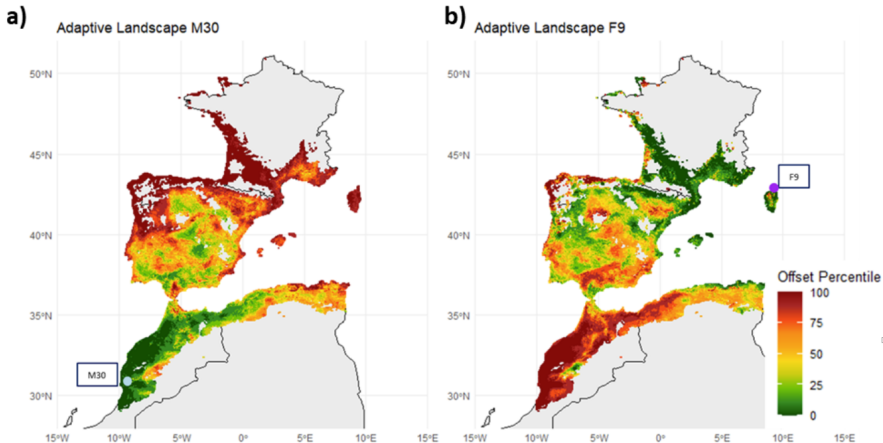
Rocchetti et al, in prep

- ENM to estimate ecological niche for restraining adaptive landscape and spatial GO generalization

ANOTHER FIELD, THE OLIVE TREE

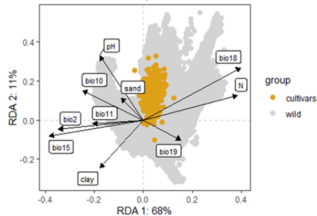


ANOTHER FIELD, THE OLIVE TREE

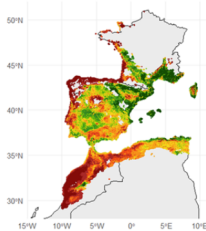


ANOTHER FIELD, THE OLIVE TREE

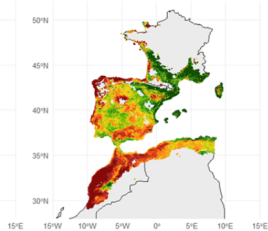
a) RDA Cultivar and Spatial Pixel Predictions



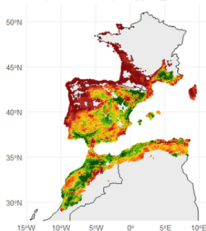
b) Adaptive Landscape Leccino



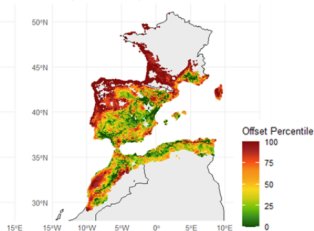
Adaptive Landscape Razaio



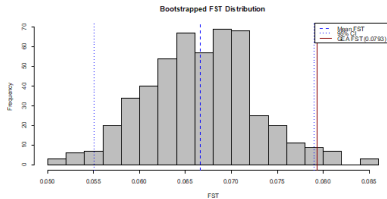
Adaptive Landscape Aggezi_Akse



Adaptive Landscape Beladi-577



- Seems to work, but not sure why



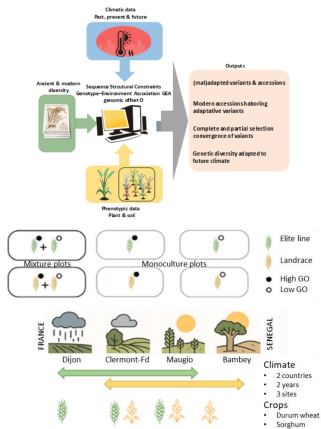
- Needs to more deeply characterize introgression in cultivated varieties

WHERE ARE WE HEADING NEXT ?

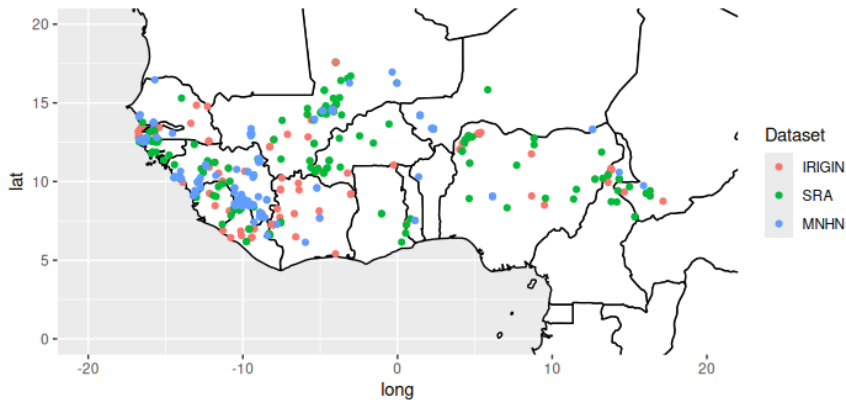
- Changing scales
 - Regional vs continental scales
- Exploring farmers perception of alloigenous varieties/landraces - implication for migration scenarios
- Are variants contributing the most to GO also enriched in deleterious alleles?
- How can we better integrate the wild relatives in the GO framework / introgression

THE PAST2Eco PROJECT - UNDER EVALUATION

- WGS on inbreds vs pool-seq on populations
- Make use of other kind of markers (kmers - genome graphs)
- Prediction of optimal environment for elite varieties
- GO and varietal mixture
- GO from museo-genomics (herbarias) to assess adaptive value lost by modern selection



WHERE ARE WE HEADING NEXT ?





- Y. Vigouroux
- A. Barnaud
- N. Scarcelli
- A.-C. Thuillet
- V. Poncet
- J. Orjuela

- A. Faye
- C. Leclerc and A. Gilabert
- O. François and T. Capblancq
- B. Sultan
- A. Assoumane
- T. Vi
- B. Rhoné



- Field experiments (African rice, Fonio, Pearl Millet)
- Methods development
- M2 **J. Lokossou** (African rice)



- Link GO / deleterious variants
- Wild compartment
- Link with social sciences
- Link with climate scientists
- Post-doc **T. Kaczmarek** (Pearl Millet, fonio)
- **A. Ilmi Ahmed** (Research engineer climate)



EWABELT

- Data generation, field trials (Fonio, Finger Millet)
- PhD **T. Kackzmarek** (fonio)
- Post-doc **M. Veltmann** (Finger Millet, co-funded AfrAdapt)



CLIM
OLIVE
MED



agropolis fondation

ClimOliveMed

- Olive tree
- PhD **L. Zunino** (GEA wild olive trees)
- Post-doc **L. Rocchetti** (GO/SDM)



BNP PARIBAS
FOUNDATION

FruitRescue

- Apple, Apricot, Grapevine, **Olive**, Peach
- M2 **A. Fall** (Olive tree)

